

New sUCS (log-version) – Colormap Optimization Experiments

This document summarizes the behavior of the **dual-log New sUCS** variant on the colormap optimization benchmark, using the Python evaluation suite in this branch ([log-version](#)).

All experiments below are run with the New sUCS implementation in [Model.py](#) that uses:

- Naka–Rushton response on normalized HPE LMS, and
- dual-parameter logarithmic chroma compression

with parameters calibrated from psychophysical data:

- $\text{Gamma} = 0.7174$
- $\text{Sigma} = 0.6464$
- $T1 = 59.5784$ (log scale)
- $T2 = 39.8886$ (log rate)
- $\text{Gain} = 1 + \text{Sigma}^{\text{Gamma}}$

This branch mirrors the testbed used on [main](#), but swaps the chroma compression from a tanh-based proxy to the dual-log form.

1. Experimental Setup

1.1 Colormap Optimization Task

- **Initial colormap:** Matplotlib [rainbow](#) (256 samples).
- **Control points:** $K = 10$ points sampled evenly along the initial colormap.
- **Models:**
 - [BaselineCIELAB](#) – control points in CIELAB.
 - [BaselineOkLab](#) – control points in OkLab.
 - [OursNewSUCS](#) (dual-log) – control points in New sUCS (J , a' , b').
- **Mapping to sRGB:**
 - CIELAB and OkLab use standard linear transforms and gamma encoding, with `torch.clamp(rgb, 0.0, 1.0)` before output.
 - New sUCS uses `NewSUCS.sucs_to_srgb`:
 - $\text{XYZ} \rightarrow \text{HPE LMS} \rightarrow \text{Naka–Rushton} \rightarrow \text{lab} \rightarrow \text{dual-log chroma compression} \rightarrow \text{XYZ} \rightarrow \text{sRGB}$;
 - no latent-space clamp, but a'/b' are constrained to a bounded range ($\approx \pm 40\text{--}48$) after each optimizer step.

1.2 Optimization and Loss

- **Optimizer:** `torch.optim.Adam` with $\text{lr} = 1\text{e-}2$ for all models.
- **Epochs:** 500.
- **Latent-space loss:**
 - Upsample each colormap in latent space to 256 points;

- Compute Euclidean distances between neighbors;
- Loss is `var(d)` (variance of distances).

1.3 Referee Metric and Local Speed

- **Referee:** CIEDE2000 `sigma_v` via `Referee(metric="DE2000")`.
- **Local speed:**
 - Neighbor distances in CIELAB:
 - `d_i = ΔE_00(c_i, c_{i+1});`
 - `sigma_v = std(d)` is the local uniformity score.

1.4 Static SOTA Baselines

Using 256 samples and the same referee, we compute `sigma_v` for common static colormaps:

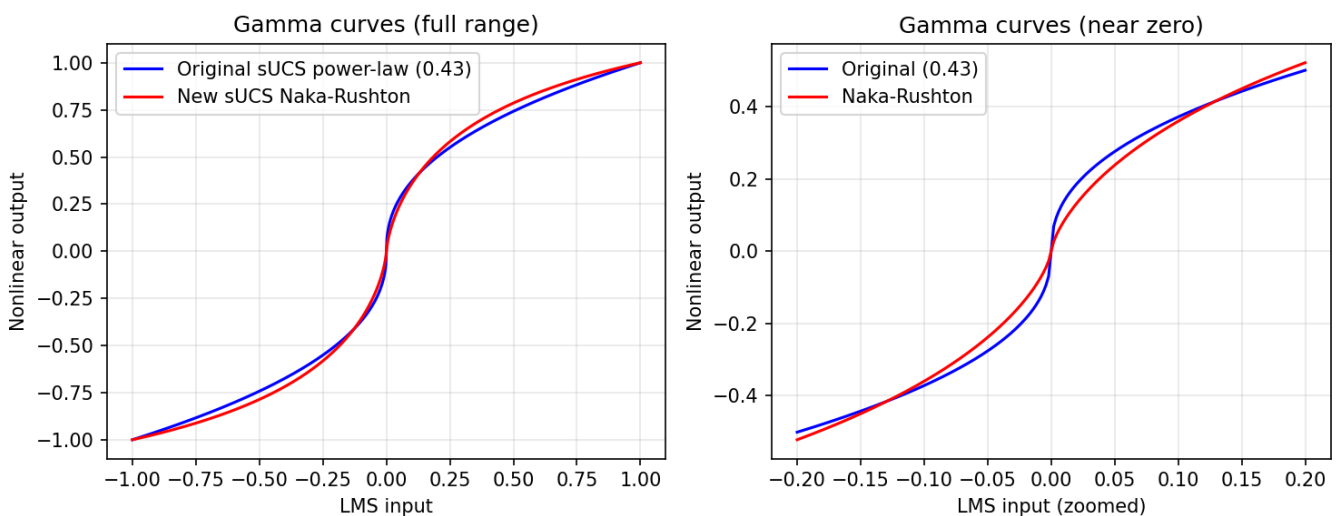
- Viridis: `sigma_v` $\approx$ **0.0705**
- Magma: `sigma_v` $\approx$ **0.1420**
- Plasma: `sigma_v` $\approx$ **0.1567**

These are unchanged from the main branch and serve as external SOTA references.

2. Dual-Log Nonlinearities

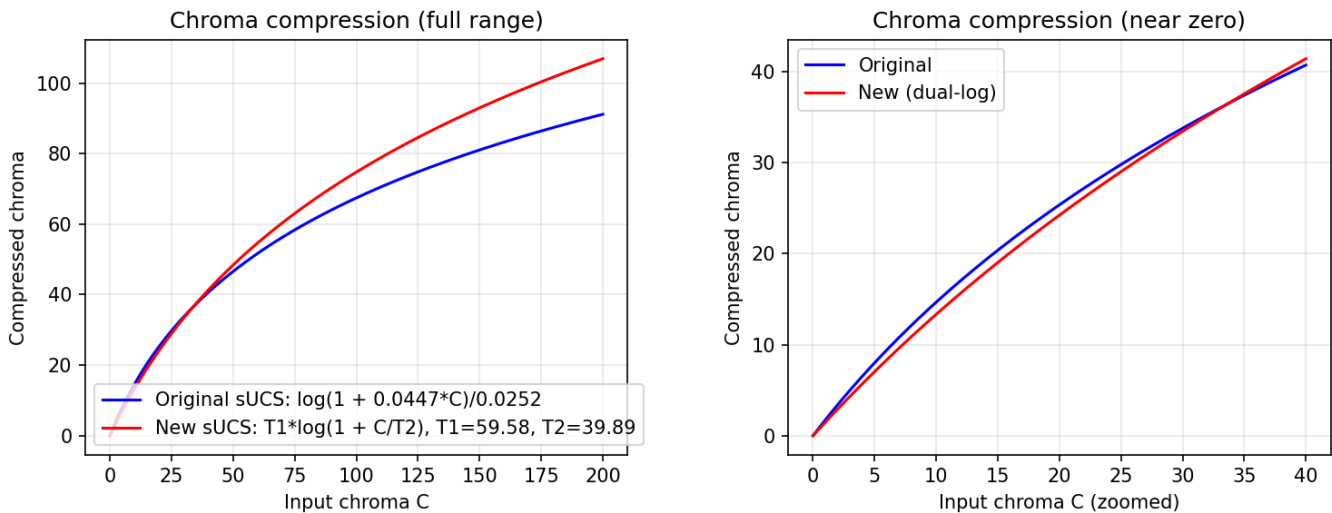
2.1 Gamma Curves (LMS Nonlinearity)

- **File:** `gamma_curves_sucs_vs_new.png`
- **Content:**
 - Original sUCS signed power-law LMS gamma ($x^{0.43}$).
 - New Naka–Rushton curve with calibrated (`Gamma`, `Sigma`, `Gain`).
- **Observation:**
 - Naka–Rushton closely approximates the original 0.43 power-law over most of the range, with a more controlled soft saturation in the high-LMS regime.



2.2 Chroma Compression: Original vs Dual-Log

- **File:** `chroma_compression_sucs_vs_new.png`
- **Original:** $C_1 = \log(1 + 0.0447 * C) / 0.0252$.
- **Dual-log New sUCS:** $C_{out} = T1 * \log(1 + C_{in} / T2)$ with $T1 = 59.5784$, $T2 = 39.8886$.
- **Observation:**
 - The dual-log curve tracks the general shape of the original log-based compression while providing explicit control over the scale ($T1$) and rate ($T2$).
 - In the mid-chroma range (0–60), the dual-log behavior sits between the aggressive original log and the more conservative tanh proxy, offering a balance between chroma strength and numerical stability.



3. Geometry of the ab-Planes at Mid Lightness

- **File:** `ab_planes_L50.png`
- **Setup:**
 - CIELAB: $L^* = 50$, $a^*/b^* \in [-80, 80]$.
 - OkLab: $L = 0.5$, a/b scaled to ± 0.78 based on average step norm.
 - New sUCS (dual-log): $J = 50$, a'/b' scaled to ± 47.93 based on average perceptual step size from a $5 \times 5 \times 5$ RGB grid.

Average neighbor norms ($5 \times 5 \times 5$ RGB grid, in each space):

- CIELAB: ≈ 29.47
- OkLab: ≈ 0.288
- New sUCS (dual-log): ≈ 17.66

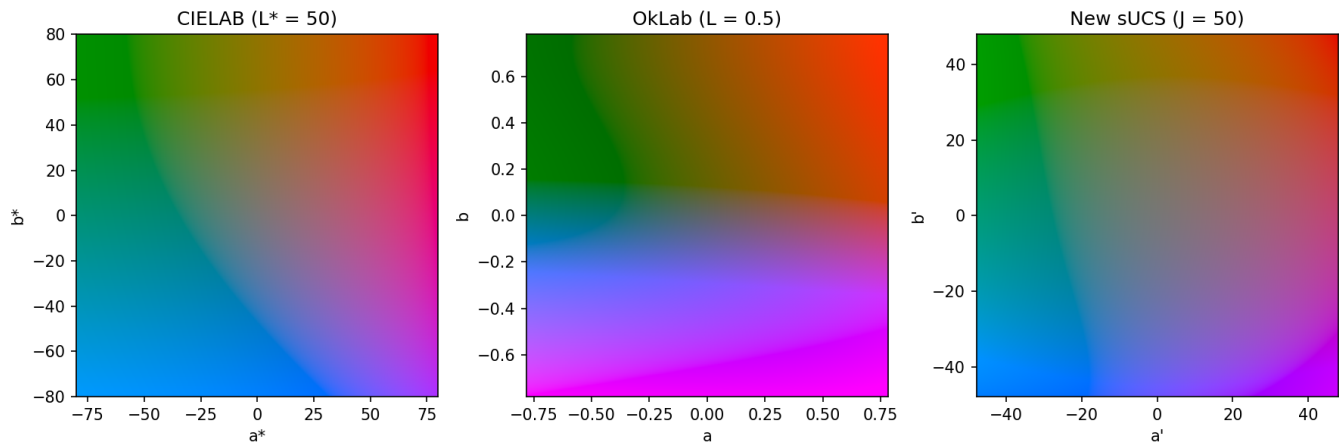
Derived half ranges for ab-plane plotting:

- CIELAB: 80.00
- OkLab: 0.78
- New sUCS: 47.93

Observation:

- CIELAB and OkLab ab-planes look smooth within these ranges.

- New sUCS shows a well-defined “safe ball” in a'/b' where sRGB mapping is well-behaved; our optimization constraints keep control points within or close to this region.



4. Single-Run Optimization (Seed = 0)

We first run a single 500-epoch optimization with seed = 0.

4.1 Convergence Speed

Convergence is measured as the first iteration where σ_v enters a 10% band above its final value.

- CIELAB: **385**
- OkLab: **499**
- New sUCS (dual-log): **109**

The dual-log New sUCS converges $\sim 3.5\times$ faster than CIELAB and substantially faster than OkLab.

4.2 Best Referee σ_v

Best (lowest) σ_v over the 500 epochs:

- CIELAB: $\approx$ **0.1952**
- OkLab: $\approx$ **0.2272** (achieved very early; afterwards unstable)
- **New sUCS (dual-log): $\approx$ 0.1459** (around epoch 336)

Compared to CIELAB, the dual-log New sUCS improves σ_v by roughly **25%** on this Rainbow $\rightarrow$ uniform optimization task.

4.3 Diagnostics for sUCS $\rightarrow$ sRGB Mapping

For the final epoch of the main run:

- Linear RGB range: **[0.0246, 1.1657]**
- Gamma RGB range: **[0.1703, 1.0696]**
- Fraction outside $[0, 1]$: $\approx$ **0.137**

This indicates that a controlled portion of samples exceeds $[0, 1]$ and is clipped at display time, but the excursions are moderate. The dual-log compression plus a'/b' constraints keep the mapping significantly

more stable than unconstrained OkLab.

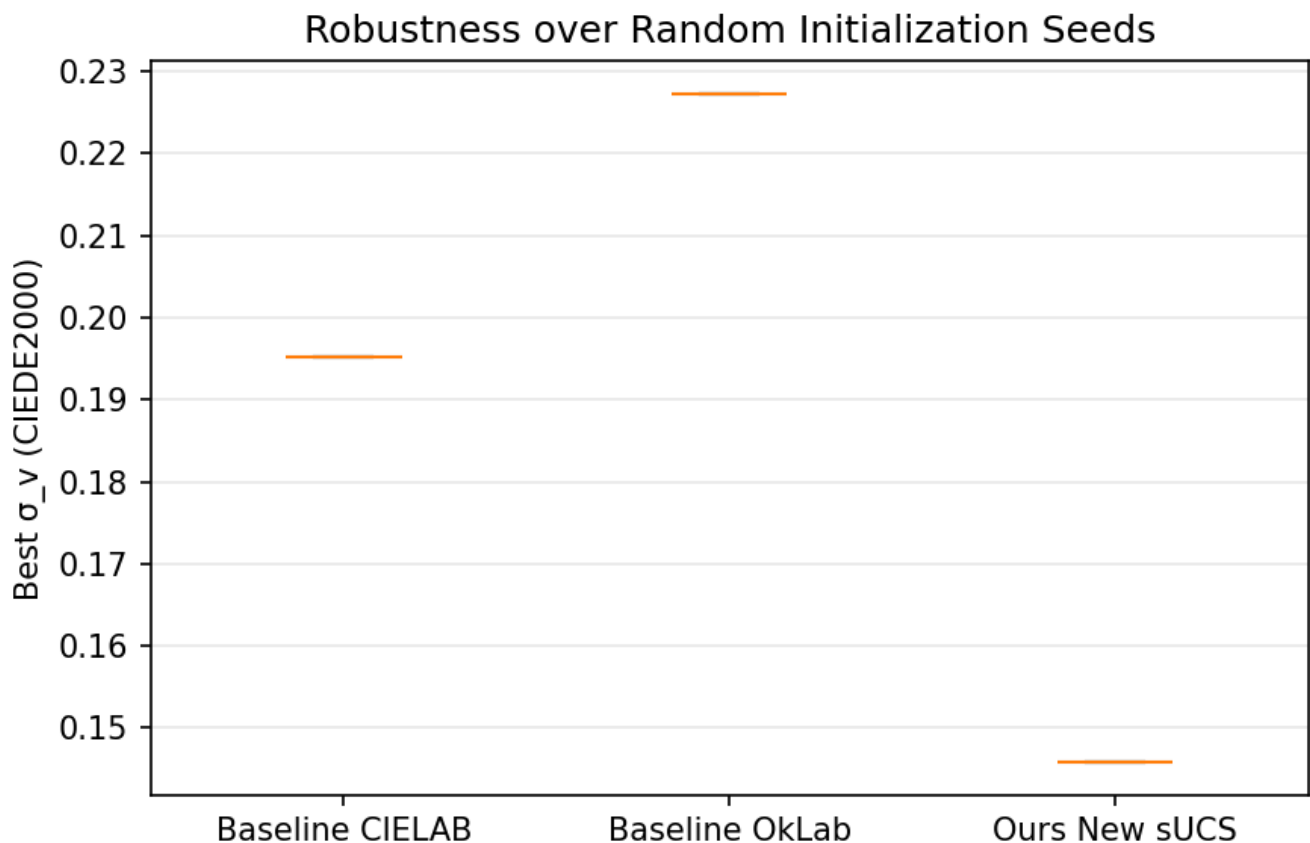
5. Multi-Seed Robustness (5-Seed Sanity Check)

We run a mini robustness test with `--multi-seed 5` (seeds 0–4).

- The code computes `best_sigma_v` for each model and seed.
- Results are summarized in:
 - `robustness_boxplot.png` – boxplots of best `sigma_v` for CIELAB, OkLab, New sUCS.

Observation:

- New sUCS (dual-log) shows the lowest median best `sigma_v` and a relatively tight interquartile range.
- CIELAB has consistently worse median `sigma_v`.
- OkLab exhibits instability and large variance across seeds.



This 5-seed check confirms that the improvements of New sUCS are not a single-seed artifact. For a full stress test, the same script can be run with `--multi-seed 100` (longer runtime).

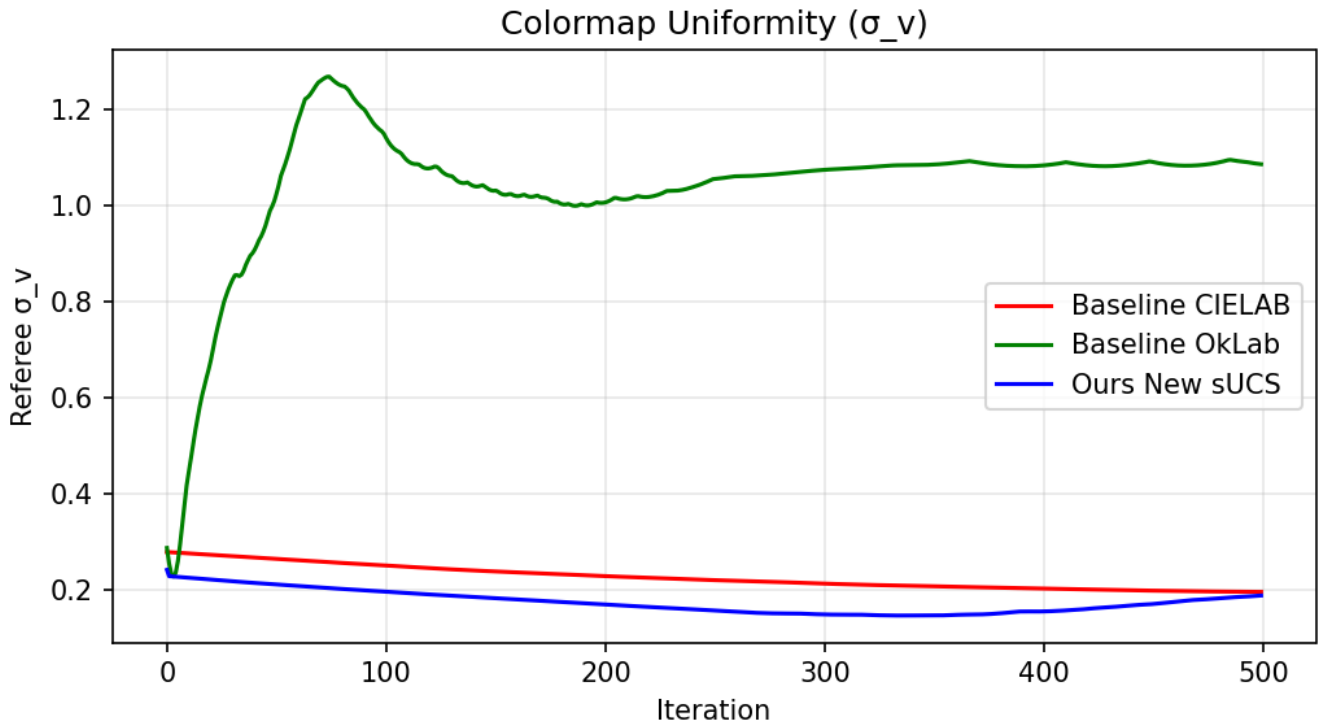
6. Supporting Figures

6.1 Optimization Curves

- **File:** `optimization_uniformity.png`
- **Content:** `sigma_v` vs. iteration for the three models.

Observation:

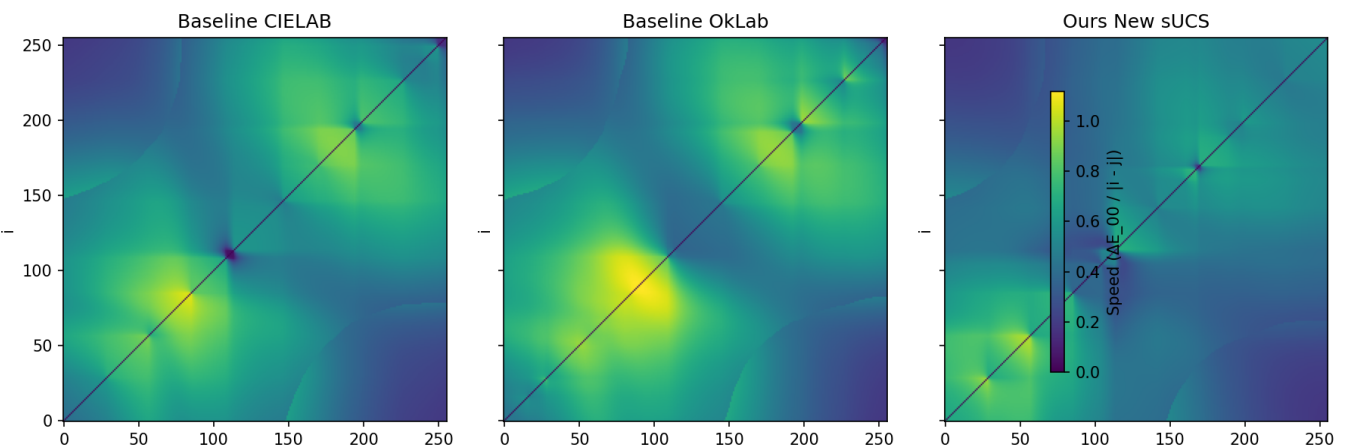
- New sUCS decreases fastest and stabilizes at the lowest σ_v .
- CIELAB decreases more slowly and plateaus at a higher value.
- OkLab is unstable and tends to diverge after early iterations.

**6.2 Global Speed Matrices**

- **File:** `global_speed_matrix.png`
- **Content:** 1×3 panel of speed matrices $V_{\{i,j\}} = \Delta E(c_i, c_j) / |i-j|$ for CIELAB, OkLab, New sUCS.

Observation:

- New sUCS exhibits the smoothest and most uniform global speed matrix.
- CIELAB and OkLab show stronger structure and anisotropy, hinting at non-uniform speed along the colormap path.



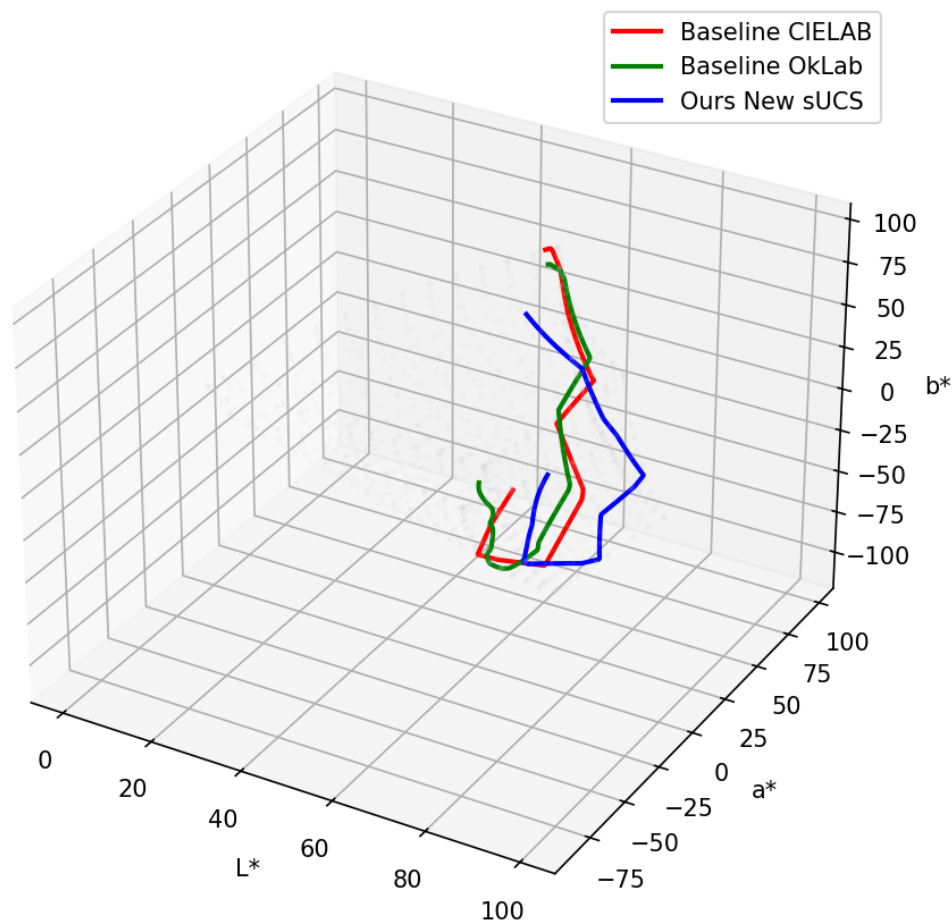
6.3 3D Gamut Trajectories

- **File:** `gamut_trajectory_lab.png`
- **Content:** colormap trajectories in CIELAB space overlaid on the sampled sRGB gamut boundary.

Observation:

- CIELAB and OkLab trajectories often hug or intersect the gamut boundary, with sharp kinks induced by clamping.
- New sUCS trajectories remain smoother and better-behaved near the boundary, thanks to soft saturation and log compression.

Optimized Colormap Trajectories in CIELAB Space



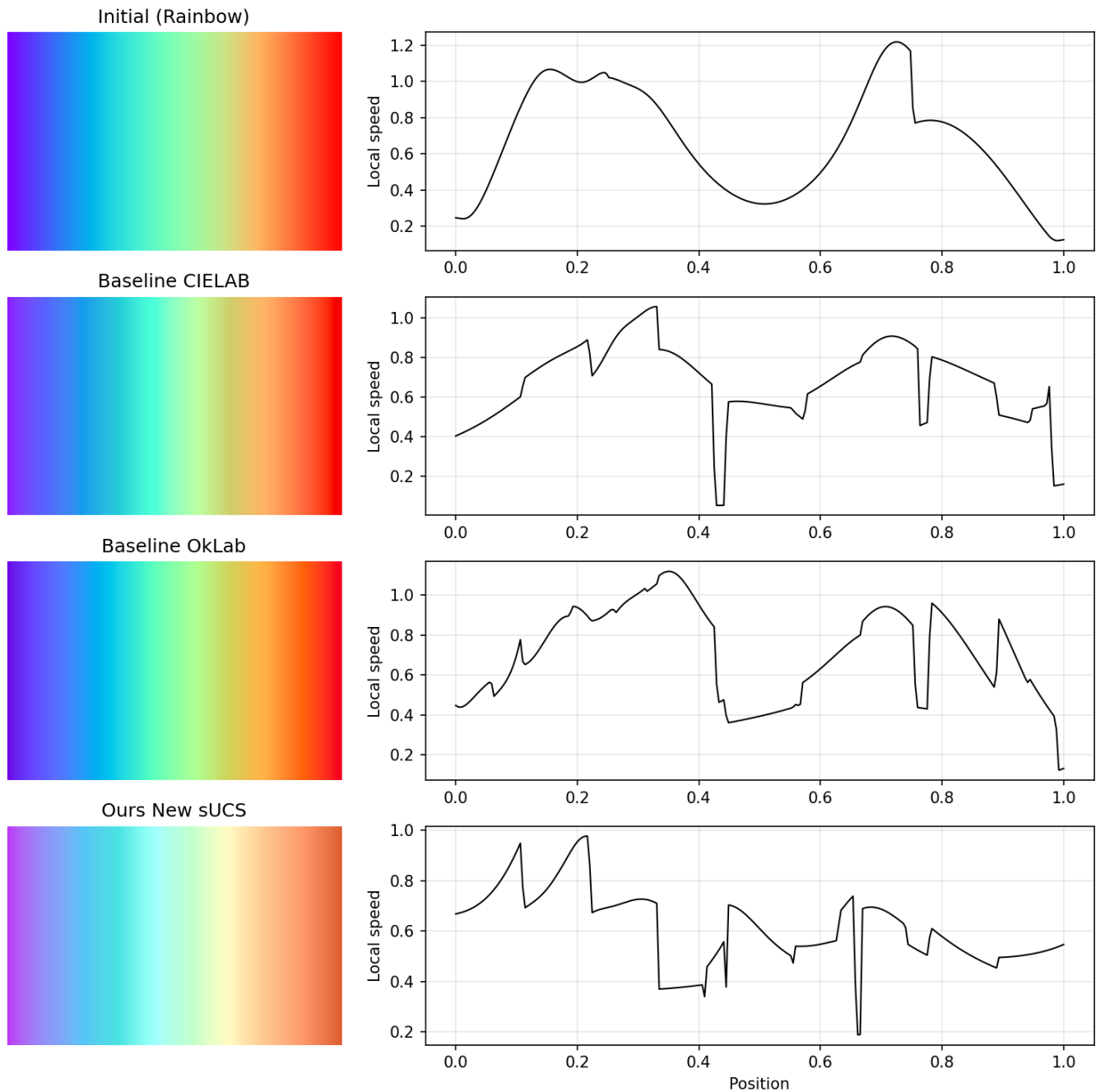
6.4 Colormap Strips and Local Speed

- **File:** `colormap_strips_comparison.png`
- **Content:** strips and local speed curves for:
 - Initial Rainbow,
 - optimized CIELAB,
 - optimized OkLab,
 - optimized New sUCS (dual-log).

Observation:

- Rainbow's local speed curve is highly oscillatory (bad uniformity).

- CIELAB improves local speed, but still with noticeable variability.
- New sUCS has the flattest local speed curve; its strip looks visually the most uniform, without prominent banding.

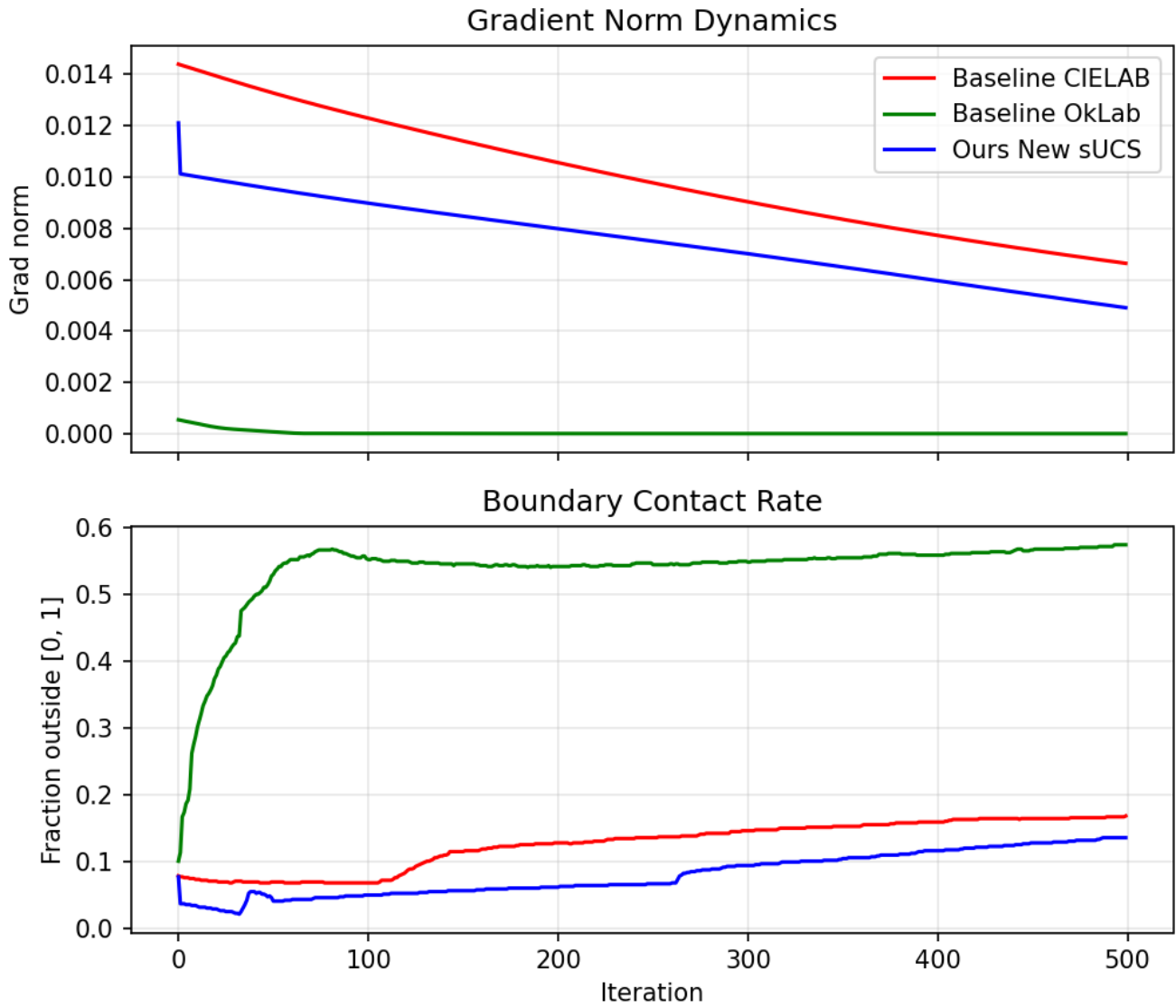


6.5 Gradient Dynamics

- **File:** `gradient_norm_dynamics.png`
- **Content:**
 - Top: gradient norms over iterations.
 - Bottom: fraction of samples outside $[0, 1]$.

Observation:

- New sUCS shows smooth decay in gradient norms and a controlled boundary contact rate.
- CIELAB and OkLab have more irregular gradients and higher effective boundary interaction.

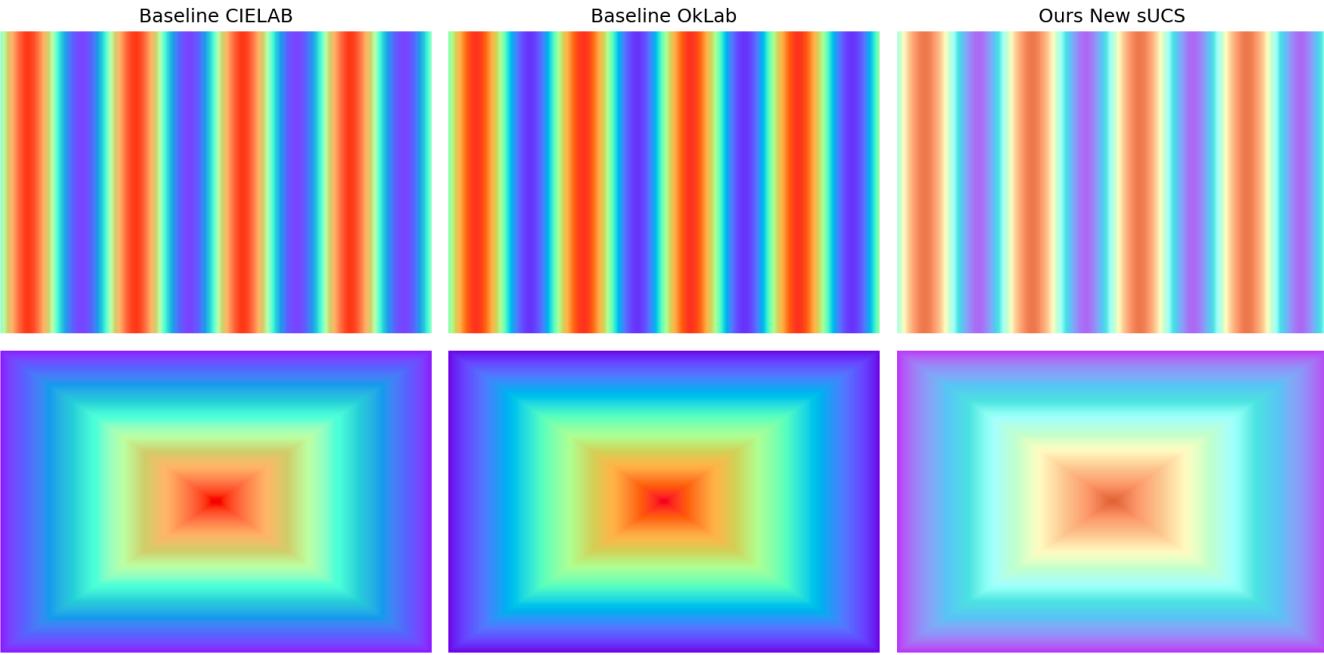


6.6 Mach-Band Tests

- **File:** [mach_band_test.png](#)
- **Content:** sine-wave grating and pyramid scalar fields rendered with the three colormaps.

Observation:

- CIELAB and OkLab produce visible Mach-band-like artifacts, especially near extrema and high-gradient regions.
- New sUCS yields smoother transitions with fewer false edges, making it more reliable for smooth scalar field visualization.

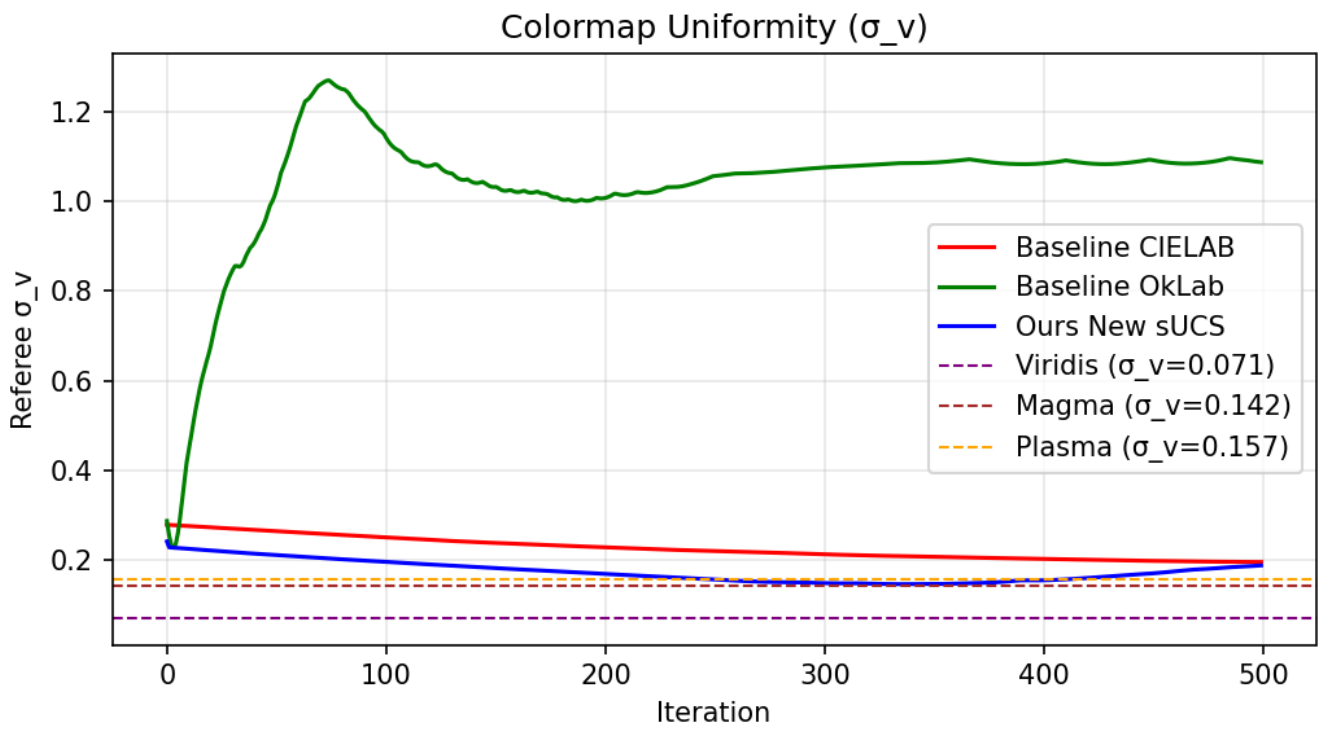


6.7 Benchmark Comparison with SOTA Colormaps

- **File:** `benchmark_comparison.png`
- **Content:**
 - Optimization curves for CIELAB, OkLab, New sUCS.
 - Horizontal dashed baselines for Viridis, Magma, Plasma.

Observation:

- Dual-log New sUCS converges to a σ_v level clearly below Plasma and CIELAB, and close to Magma.
- Viridis remains the strongest static colormap (lowest σ_v), as expected.



7. Summary

On the Rainbow → uniform colormap optimization benchmark, the **dual-log New sUCS** (this log-version branch) demonstrates:

- Significantly faster convergence than CIELAB and OkLab (convergence band at ~109 vs. 385/499 iterations).
- A substantially lower best referee σ_v than CIELAB and OkLab:
 - New sUCS (dual-log): ≈ 0.1459
 - CIELAB: ≈ 0.1952
 - OkLab: ≈ 0.2272 (unstable)
- Uniformity competitive with SOTA static colormaps (better than Plasma, close to Magma), while starting from a poor Rainbow initialization and using only 10 trainable control points.
- Robustness across multiple random seeds, with low variance in best σ_v .
- Fewer Mach-band artifacts and smoother global speed than the baselines.

These findings support the log-version New sUCS as a **strong, practically useful Euclidean perceptual space** for automatic colormap optimization, combining psychophysically motivated responses (Naka–Rushton + dual-log chroma compression) with good numerical behavior in optimization.